
Riyad Cornu

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Education

Toulouse INP – ENSEEIHT (N7) <i>Engineering degree, Systems & Networks Architecture (ASR) track</i>	Toulouse, France 2025 – 2028 (expected)
• Core curriculum: computer architecture, networks & telecommunications, language translation / compilation, concurrent & distributed systems, formal methods (Why3, Coq).	
CPGE MPI, Lycée Paul Valéry <i>Preparatory classes — Mathematics, Physics, Computer Science</i>	Paris, France 2024 – 2025
CPGE MP2I-MPI, Lycée Hoche <i>Preparatory classes — Mathematics, Physics, Computer Science</i>	Versailles, France 2022 – 2024

Experience

Basic Fit <i>Front Desk Agent</i>	Vélizy, France 06/2024 – 09/2024 and 06/2025 – 09/2025
• Managed customer relations, membership sales, and after-sales support. • Resolved conflicts and ensured proper functioning of the club facilities.	
Auchan Drive <i>Multi-skilled Team Member</i>	Vélizy, France 02/2022 – 07/2022
• Prepared customer orders and managed stock and inventory levels.	

Projects

Neural Network from Scratch	2025
• Built a neural network from scratch in CUDA, with no ML framework, including hand-written GPU kernels for every matrix operation. • Implemented backpropagation and the Adam optimizer; reached 96% accuracy on MNIST and 89% on EMNIST.	
Unix Mini-shell	2026
• Unix shell in C: multi-command pipelines, I/O redirection (</>), background execution (&), and builtins (cd, exit, rep). • Process management via fork/execvp and pipes (pipe/dup2); asynchronous synchronization through a SIGCHLD handler (sigaction) and sigsuspend, with no busy-waiting or zombie processes.	
Memory Allocator (malloc)	2026
• Reimplemented malloc/free/calloc/realloc in C directly on top of Linux primitives (mmap, sysconf), with no dependency on libc. • Linked-list block allocator with splitting and reuse of freed regions; validated with a unit test suite (memory alignment, multiple allocations, calloc, realloc, edge cases).	
CHIP-8 Emulator	2026
• Wrote a CHIP-8 interpreter in C, implementing the full instruction set (opcode decoding, timers, stack, registers). • Used SDL2 for display rendering and keyboard input; validated against the IBM logo, Corax+, and BC_test ROMs.	

Certifications

Project Management MOOC (GdP27)	March 2026
• Project Management MOOC (Rémi Bachelet, Centrale Lille), 27th session — specialization modules "MRP" (Problem-Solving Methodology) and "GPAS" (Agile Project Management with Scrum) — mooc.gestiondeprojet.pm	

Skills & Interests

Technical: C, Rust, Ada, CUDA (systems & low-level); Python, MATLAB (applied/scientific computing); Why3, Coq (formal methods)

Languages: French (native), English (operational)

Interests: Strength training & powerlifting, reading